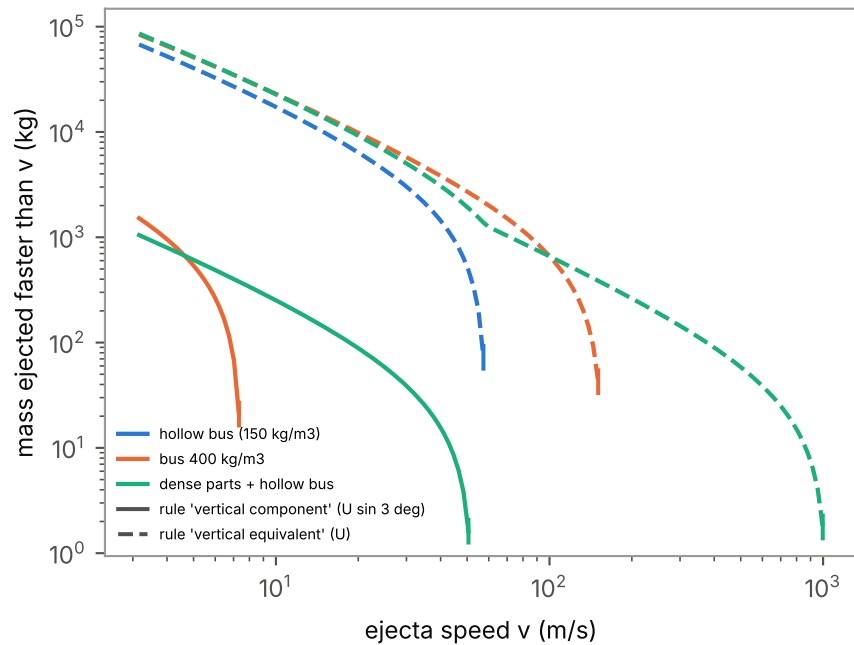
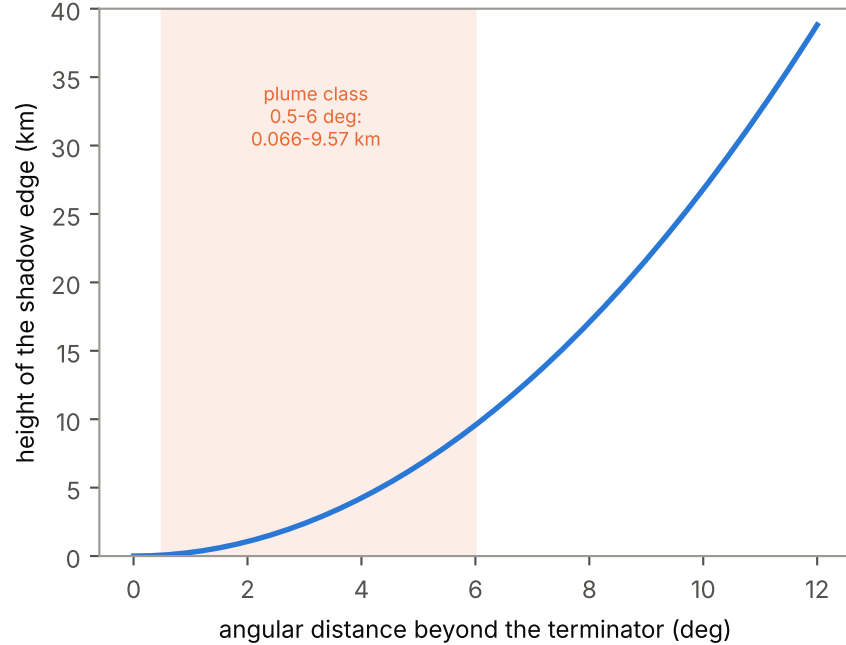


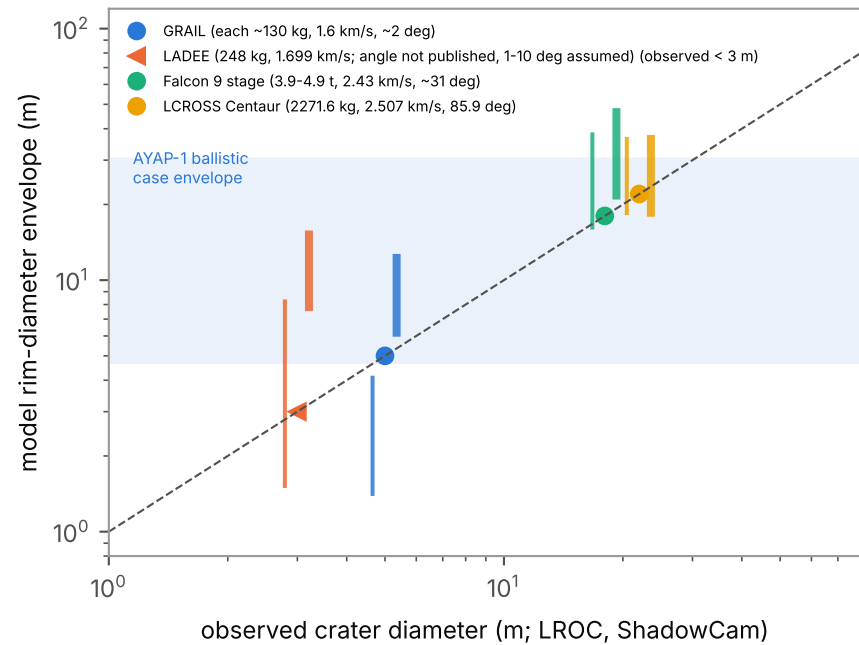
a) Ejecta mass-speed, ballistic case (HH2011, sand); curves stop at the scaling-domain edge (bar): faster = cannot determine



b) Height ejecta must reach to be sunlit (point Sun, sphere; the finite solar disc and terrain broaden the edge)



c) Qualitative consistency, no tuning (thin: vertical-component rule; thick: vertical-equivalent; parameter sets x densities 150-1000)



MODEL; panel c compares with OBSERVED crater sizes (research/precedents.csv); broad envelopes, not a validation of either oblique-impact rule